WORKSHEET

Coefficient of determination

1 The regression line:

$$\textit{Mark in test} = 7.8 + 12.3 \times \textit{Number of hours studied}$$

has a coefficient of determination of 0.66.

a Interpret the:

i gradient

ii y-intercept

iii coefficient of determination.

b State whether the interpretations of gradient and y-intercept are relevant.

2 The least-squares regression line that determines a household heating bill for the winter by the average temperature is:

$$\textit{Heating bill (\$)} = 1428 - 56.9 \times \textit{Average outdoor temperature}$$

The coefficient of determination for these data is 0.77.

a Interpret the:

i gradient

ii y-intercept

iii coefficient of determination.

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b State whether the interpretations of gradient and y-intercept are relevant.

3 The regression line:

$$\textit{Muscle gain (kg)} = 0.013 \times \textit{Number of hours of exercise}$$

has a coefficient of determination of 0.74.

a Interpret the:

i gradient

ii y-intercept

iii coefficient of determination.

b State whether the interpretations of gradient and y-intercept are relevant.

4 The regression line:

$$\textit{Maximum sign legibility distance (m)} = 175.9 - 0.91 \times \textit{Age of driver}$$

has a coefficient of determination of 0.76.

a Interpret the:

i gradient

ii y-intercept

iii coefficient of determination.

b State whether the interpretations of gradient and y-intercept are relevant.

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5 The least-squares regression line that determines a person's hand span using their height is:

$$\textit{Hand span (cm)} = -3 + 0.14 \times \textit{Height (cm)}$$

The coefficient of determination for these data is 0.83.

a Interpret the:

i gradient

ii y-intercept

iii coefficient of determination.

b State whether the interpretations of gradient and y-intercept are relevant.

6 The regression line that determines the number of eggs laid per week by a hen by the age of the hen in months is:

$$\textit{Number of eggs laid per week} = 7 - 0.15 \times \textit{Age of hen (months)}$$

Note: This regression line is only relevant for hens 5 months or older. The coefficient of determination for these data is 0.89.

a Interpret the:

i gradient

ii y-intercept

iii coefficient of determination.

b State whether the interpretations of gradient and y-intercept are relevant.

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Answers

1 a i The mark in test increases by 12.3 for every hour increase in study time.

ii The mark in test is 7.8 when hours of study is zero.

iii 66% of variation in marks in the test can be explained by the variation in the number of hours of study.

b The interpretations of gradient and y-intercept are both relevant.

2 a i The heating bill decreases by \$56.90 for every one degree increase in the average outdoor temperature.

ii The heating bill is \$1428 when the average outdoor temperature is zero.

iii 77% of the variation in the heating bill can be explained by the variation in the average outdoor temperature.

b The interpretations of gradient and y-intercept are both relevant.

3 a i Muscle gain increases by 0.013 kg for every hour increase in exercise.

ii Muscle gain is zero when hours of exercise are zero.

iii 74% of variation in muscle gain can be explained by variation in the number of hours of exercise.

b The interpretations of gradient and y-intercept are both relevant.

4 a i Maximum sign legibility distance decreases by 0.91 m for every 1 year increase in the age of the driver.

ii Maximum sign legibility distance is 175.9 m when age of driver is 0.

iii 76% of the variation in the maximum sign legibility distance can be explained by variation in the age of the driver.

b The gradient is relevant. The y-intercept is not relevant, as infants at the age of zero cannot drive or read road signs.

5 a i A person's hand span increases by 0.14 cm for every 1 cm increase in their height.

ii A person's hand span is -3 cm when their height is 0.

iii 83% of the variation in hand span can be explained by the variation in the person's height.

b Gradient is relevant. The y-intercept is not relevant, as a person's hand span can't be negative.

6 a i The number of eggs laid decreases by 0.15 for every 1 month increase in the age of the hen.

ii The number of eggs laid per week is 7 when the age of the hen in months is zero.

iii 89% of the variation in the number of eggs laid per week can be explained by the variation in the hen's age.

b The interpretation of the gradient is relevant although a hen cannot lay 0.15 of an egg, but the y-intercept interpretation is not relevant because newborn and young chicks do not lay eggs.

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